MOTS-C

Exercise 6–10 mg SC 2–3× weekly; healthspan 3–6 mg SC 1–2× weekly

MOTS-c has demonstrated metabolic, exercise-capacity, and pancreatic-islet effects in specific mouse models, supporting plausible applications to human metabolic health and physical function. [1,2,4,8,9] MOTS-c is a 16-amino-acid peptide encoded within the mitochondrial 12S ribosomal RNA gene. [1] It influences folate and purine metabolism, AICAR accumulation, and AMPK signaling in experimental systems. [1] Under metabolic stress, it can translocate to the nucleus and regulate protective gene expression, including NRF2-associated responses. [3] These actions support its investigation as a partial exercise mimetic. [2,3]

💉 Human safety is plausible, not established. [5]

⭐ ⭐ Demonstrated metabolic and physical-capacity effects in mice; plausible corresponding human benefits. [1,2,4]

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

MOTS-c improved insulin sensitivity and physical capacity in mouse experiments. [1,2] A small mouse crossover experiment also found improved running after a single injection, supporting a possible rapid performance effect. [4]

Dosage & Patterns

The following speculative, experimental SC patterns are community-informed examples:

- **Exercise-mimetic pattern**: 6–10 mg SC, 2–3 times weekly on rest days.
- **Healthspan pattern**: 3–6 mg SC, 1–2 times weekly on rest days.
- **Acute pre-exercise**: Administration 30–60 minutes before training is another speculative timing approach. An acute pre-exercise benefit was observed in mice, so rapid action is biologically possible; its relevance to human SC timing remains an extrapolation, however the human equivalent dose would be approximately 80 mg. [4]

Combining AMPK activators requires individual assessment

Combining metabolic interventions may modify exercise adaptation. In a human trial, metformin attenuated selected aerobic-training adaptations, with substantial individual variability. [6] An additive AMPK-related interaction with MOTS-c is a mechanistic hypothesis. [1,3,6] Potential overlapping influences include:

- Metformin and proposed AMPK-active compounds such as AICAR or berberine; these are mechanistically distinct interventions, not interchangeable measures of AMPK activity. [1,6,14]
- Frequent hard training or prolonged fasting, which can alter metabolic demand and stress responses. [2,3]
- Cold exposure, considered here as a possible additional metabolic demand.

MOTS-c activates AMPK in experimental systems. [1] Combining these interventions could plausibly alter adaptation or recovery.

Benefits

- **Improved insulin sensitivity** in mice: reduced diet- and age-associated insulin resistance supports a plausible human metabolic benefit. [1]
- **Plausible metabolic flexibility**: altered fuel utilization in cells and mice supports this possibility. [1,2]
- **Improved exercise capacity** in mice: convergent running-performance findings support a plausible human endurance benefit. [2,4]
- **Possible acute exercise-capacity benefit**: a single high dose injection improved running performance in a six-mouse crossover experiment. [4]
- **Preservation of physical function** in old mice: late-life treatment improved functional measures, supporting a plausible human healthspan application. [2]
- **Pancreatic-islet protection** in experimental diabetes: mouse studies show reduced autoimmune islet destruction and senescence-associated changes, supporting a plausible human application. [8,9]
- **Reduced diet-induced fat gain** in mice: protection against high-fat-diet obesity supports a plausible body-composition application. [1]
- **Plausible cellular stress resilience**: protective gene regulation and improved cell survival under metabolic stress support this interpretation. [2,3]

Subjective practitioner reports include steadier energy, less exertion, improved mood, faster recovery, and less soreness. These are anecdotal impressions; the personal account below records an opposite exercise response.

Investigated Uses

- **Insulin resistance and metabolic dysfunction:** preclinical research targets. [1,8]
- **Age-related physical decline and metabolic stress:** preclinical targets; fatigue alone does not diagnose mitochondrial dysfunction. [2,3]

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Clinical Cautions

- **Shared metabolic signaling** provides a plausible reason to assess combinations individually. [1,3,6]
- **Pregnancy and breastfeeding:** Avoid experimental use because reproductive and developmental safety has not been established. [5]

Drivers of Diminished Response

MOTS-c is not established as a conventional receptor agonist. [5] Neutralizing antibodies are a plausible cause of diminished response, and downstream pathway adaptation is another mechanistic possibility. [3,5]

Biological Mechanism

MOTS-c links mitochondrial genetic information with metabolic and nuclear signaling. Cell experiments identify altered folate and purine metabolism, AICAR accumulation, and AMPK activation. [1] Under stress, MOTS-c can enter the nucleus and influence protective gene expression, including NRF2-associated responses. [3] In experimental autoimmune diabetes, it also modulated T-cell phenotype and TCR–mTORC1 signaling. [9] These are context-dependent mechanisms of metabolic and immune adaptation.

My Workout Experience and Possible Explanations

The symptoms below are personal observations. I expected MOTS-c to improve workout energy, but my experience was the opposite.

Excessive or sustained AMPK activation may impair muscle recovery by suppressing mTOR-mediated protein synthesis and muscle regeneration. Animal studies demonstrate impaired muscle repair and attenuated hypertrophy under conditions of increased AMPK signaling. Persistent activation could therefore contribute to inadequate recovery and declining physical performance, although a direct causal relationship with fatigue in humans remains unestablished. [16-18]

Training Load and Recovery

Poor performance, weakness, and fatigue can occur when training exceeds recovery. [15] My proposed explanation is that several metabolic interventions altered my balance between stress and adaptation. MOTS-c engages AMPK-related signaling. [1,3]

The Combination I Was Using

I did all the following at the same time:

- Intense, frequent workouts… increasing metabolic demands.
- A mini-cut with sustained calorie deficit… reducing available dietary energy.

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- Metformin for healthspan… potentially modifying metabolic and exercise responses. [6]
- And then I added MOTS-c… another intervention affecting metabolic signaling. [1,3]

The Results Were Ugly

During my workout, I had to quit halfway through. I simply had no energy. An hour later, I bent down to pick something up and straightened up and got lightheaded and dizzy.

What This Experience Can Guide

I speculate that my longstanding exercise, diet, and metformin regimen had already created substantial metabolic demands and AMPK activation, and that adding MOTS-c contributed to an unfavorable response. An AMPK-related interaction remains a proposed mechanism for my experience.

Emerging Research Around Acute Exercise

Hyatt's crossover experiment in six female mice used approximately 83 mg HED IP as a single injection 10 minutes before running. Distance and time increased by approximately 15% and 12%, respectively. The mechanism remains uncertain. [4,11]

Emerging Research Around Pancreas Support

Islet-cell senescence: A 2025 study found reduced senescence-associated changes and improved glucose handling in mouse diabetes models. Its human component compared circulating MOTS-c levels. The findings support a plausible human islet-protective application. [8]

Conclusions

MOTS-c has an experimental basis for metabolic stress adaptation and demonstrated effects in selected mouse models. [1–4,8,9] Human metabolic and physical-function applications are plausible extrapolations. "Exercise mimetic" refers to selected overlapping effects. [2]

Community dose and cycle examples remain speculative. A registered native MOTS-c trial has no posted results. [10]

Older or metabolically impaired populations are reasonable research targets. [1,2,8]

Animal performance findings also justify studying active subjects. My adverse experience is a personal tolerability observation; the proposed AMPK explanation remains speculative. [2,4]

References

[1] Lee C, Zeng J, Drew BG, et al. The mitochondrial-derived peptide MOTS-c promotes metabolic homeostasis and reduces obesity and insulin resistance. *Cell Metab*. 2015;21(3):443-454. doi:10.1016/j.cmet.2015.02.009

[2] Reynolds JC, Lai RW, Woodhead JST, et al. MOTS-c is an exercise-induced mitochondrial-encoded regulator of age-dependent physical decline and muscle homeostasis. *Nat Commun*. 2021;12(1):470. doi:10.1038/s41467-020-20790-0

[3] Kim KH, Son JM, Benayoun BA, Lee C. The Mitochondrial-Encoded Peptide MOTS-c Translocates to the Nucleus to Regulate Nuclear Gene Expression in Response to Metabolic Stress. *Cell Metab*. 2018;28(3):516-524.e7. doi:10.1016/j.cmet.2018.06.008

[4] Hyatt JPK. MOTS-c increases in skeletal muscle following long-term physical activity and improves acute exercise performance after a single dose. *Physiol Rep*. 2022;10(13):e15377. doi:10.14814/phy2.15377

[5] US Food and Drug Administration. FDA Evaluation of MOTS-c–Related Bulk Drug Substances (MOTS-c (free base) and MOTS-c acetate). Pharmacy Compounding Advisory Committee briefing document. May 11, 2026. Accessed October 23, 2026. Source

[6] Konopka AR, Laurin JL, Schoenberg HM, et al. Metformin inhibits mitochondrial adaptations to aerobic exercise training in older adults. *Aging Cell*. 2019;18(1):e12880. doi:10.1111/acel.12880

[7] CohBar, Inc. CohBar Announces Positive Topline Results from the Phase 1a/1b Study of CB4211 Under Development for NASH and Obesity. News release. August 10, 2021. Accessed October 23, 2026. Source

[8] Kong BS, Lee H, L'Yi S, Hong S, Cho YM. Mitochondrial-encoded peptide MOTS-c prevents pancreatic islet cell senescence to delay diabetes. *Exp Mol Med*. 2025;57(8):1861-1877. doi:10.1038/s12276-025-01521-1

[9] Kong BS, Min SH, Lee C, Cho YM. Mitochondrial-encoded MOTS-c prevents pancreatic islet destruction in autoimmune diabetes. *Cell Rep*. 2021;36(4):109447. doi:10.1016/j.celrep.2021.109447 Published correction: *Cell Rep*. 2021;36(11):109724. doi:10.1016/j.celrep.2021.109724

[10] Hudson Biotech. MOTS-c for Improving Insulin Sensitivity in Adults With Prediabetes and Overweight/Obesity. ClinicalTrials.gov identifier: NCT07505745. Updated April 1, 2026. Accessed October 23, 2026. Source

[11] US Food and Drug Administration. Estimating the Maximum Safe Starting Dose in Initial Clinical Trials for Therapeutics in Adult Healthy Volunteers. Guidance for industry. July 2005. Accessed October 23, 2026. Source

[12] Rice MC, Imun M, Jung SW, et al. MOTS-c is a mitochondrial-encoded interferon-linked host defense peptide. *Elife*. 2026;12:RP87615. doi:10.7554/elife.87615

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[13] Jamnick NA, Livingston PD, Gammon CJ, Weinzierl NM, Novinger LJ, Bonetto A. MOTS-c partially protects against skeletal muscle deterioration in C26 cachexia. *Front Med (Lausanne)*. 2026;13:1838178. doi:10.3389/fmed.2026.1838178

[14] Turner N, Li JY, Gosby A, et al. Berberine and its more biologically available derivative, dihydroberberine, inhibit mitochondrial respiratory complex I: a mechanism for the action of berberine to activate AMP-activated protein kinase and improve insulin action. *Diabetes*. 2008;57(5):1414-1418. doi:10.2337/db07-1552

[15] Meeusen R, Duclos M, Foster C, et al. Prevention, diagnosis, and treatment of the overtraining syndrome: joint consensus statement of the European College of Sport Science and the American College of Sports Medicine. Med Sci Sports Exerc. 2013;45(1):186-205. https://doi.org/10.1249/MSS.0b013e318279a10a

[16] LaBarge S, McDonald M, Smith-Powell L, Auwerx J, Huss JM. Estrogen-related receptor-α (ERRα) deficiency in skeletal muscle impairs regeneration in response to injury. FASEB J. 2014;28(3):1082-1097. https://doi.org/10.1096/fj.13-229211

[17] Mounier R, Lantier L, Leclerc J, et al. Important role for AMPKα1 in limiting skeletal muscle cell hypertrophy. FASEB J. 2009;23(7):2264-2273. https://doi.org/10.1096/fj.08-119057

[18] Dreyer HC, Fujita S, Cadenas JG, Chinkes DL, Volpi E, Rasmussen BB. Resistance exercise increases AMPK activity and reduces 4E-BP1 phosphorylation and protein synthesis in human skeletal muscle. J Physiol. 2006;576(Pt 2):613-624. https://doi.org/10.1113/jphysiol.2006.113175

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